

Crystal structure and catalytic mechanism of PL35 family glycosaminoglycan lyases with an ultrabroad substrate spectrum

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eLife Assessment

This **useful** manuscript reports on the crystal structures of two glycosaminoglycan (GAG) lyases from the PL35 family, along with in vitro enzyme activity assays and comprehensive structure-guided mutagenesis. The authors have addressed key concerns by incorporating additional docking analyses, validating the role of His188 in alginate degradation, and providing ICP-MS data to examine Mn^{2+} binding. While these improvements enhance the study, the study is **incomplete** due to the lack of enzyme-substrate complex structures and reliance on modeling which still limit mechanistic insight. Nonetheless, the revised manuscript presents a more complete analysis that will be of interest to specialists in carbohydrate-active enzymes.

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Abstract

Recently, a new class of glycosaminoglycan (GAG) lyases (GAGases) belonging to PL35 family has been discovered with an ultrabroad substrate spectrum that can degrade three types of uronic acid-containing GAGs (hyaluronic acid, chondroitin sulfate and heparan sulfate) or even alginate. In this study, the structures of GAGase II from *Spirosoma fluviale* and GAGase VII from *Bacteroides intestinalis* DSM 17393 were determined at 1.9 and 2.4 Å resolution, respectively, and their catalytic mechanism was investigated by the site-directed mutant of their crucial residues and molecular docking assay. Structural analysis showed that GAGase II and GAGase VII consist of an N-terminal $(\alpha/\alpha)_6$ toroid multidomain and a C-terminal two-layered β -sheet domain with Mn^{2+} . Notably, although GAGases share similar folds and catalytic mechanisms with some GAG lyases and alginate lyases, they exhibit higher

structural similarity with alginate lyases than GAG lyases, which may present a crucial structural evidence for the speculation that GAG lyases with $(\alpha/\alpha)_n$ toroid and antiparallel β -sheet structures arrived by a divergent evolution from alginate lyases with the same folds. Overall, this study not only solved the structure of PL35 GAG lyases for the first time and investigated their catalytic mechanism, especially the reason why GAGase III can additionally degrade alginate, but also provided a key clue in the divergent evolution of GAG lyases that originated from alginate lyases.

Introduction

Glycosaminoglycans (GAGs) are a class of linear polyanionic polysaccharides ubiquitously distributed on the cell surface and in the extracellular matrix (ECM) of animal tissues (Cohen and Merzendorfer, 2019 [\[1\]](#)), and they participate in various physiological and pathological processes through interacting with a series of chemokines (Dyer, et al., 2015 [\[2\]](#); Irie, et al., 2008 [\[3\]](#)), growth factors (Sirko, et al., 2010 [\[4\]](#); Zhang, et al., 2019 [\[5\]](#)) or other ECM components. Except for keratan sulfate (KS), which does not contain hexuronic acid (HexUA) residues, other GAGs are composed of repeating disaccharides consisted of HexUA (D-glucuronic acid (GlcUA) or L- iduronic acid (IdoUA)) and hexosamine (HexNAc) (D-N-acetylgalactosamine (GalNAc) or D-N- acetylglucosamine (GlcNAc)). Based on the disaccharide composition and the type of glycosidic bonds between disaccharides units, HexUA-containing GAGs are classified into three classes: hyaluronan (HA), chondroitin sulfate (CS)/dermatan sulfate (DS) and heparan sulfate (HS)/heparin (Hep) (Cohen and Merzendorfer, 2019 [\[1\]](#)). HA is the only non-sulfated GAG composed of repeating -GlcUA β 1-3GlcNAc- disaccharides linked by β 1-4 glycosidic bonds. In contrast, the structures of CS/DS and Hep/HS are highly complex due to sulfation and epimerization modifications. The backbone of CS is composed of repeating -GlcUA β 1-3GalNAc- disaccharides linked by β 1-4 glycosidic bonds, and is further sulfated at C-2 of GlcUA and C-4/C-6 of GalNAc by sulfotransferases; meanwhile, the GlcUA residues in CS can be epimerized into IdoUA residues by glucuronyl C-5 epimerase to form DS (Kusche-Gullberg and Kjellen, 2003 [\[6\]](#)). Similarly, HS/Hep composed of repeating -4GlcUA β 1-4GlcNAc- units can be sulfated at C-2 of GlcUA and N/C-2/C-3/C-6 of GlcNAc/deacetylated GlcN, and GlcUA residues are often epimerized to IdoUA residues, especially in Hep (Cohen and Merzendorfer, 2019 [\[1\]](#)). Such complex modification through sulfation and epimerization endows sulfated GAGs with diverse biological functions.

GAG lyases, belonging to the superfamily of polysaccharide lyases (PLs), specifically catalyze the degradation of HexUA-containing GAGs (Charnock, et al., 2002 [\[7\]](#); Garron and Cygler, 2010 [\[8\]](#)). These reactions generate a plethora of oligosaccharides that contain an unsaturated double bond between C-4 and C-5 on uronic acids at their nonreducing ends, which is generated via a β -elimination mechanism (Garron and Cygler, 2010 [\[8\]](#)). As a complimentary mechanistic strategy to glycoside hydrolases (GHs), GAG lyases degrade HexUA-containing GAGs without the participation of a water molecule and involved in the polysaccharide metabolism of microorganisms (Garron and Cygler, 2010 [\[8\]](#); Ndeh, et al., 2020 [\[9\]](#)). For the past many years, an increasing number of gene sequences containing PL catalytic modules and some ancillary modules have been annotated as PLs (Lombard, et al., 2010 [\[10\]](#)). Based on their sequence similarity, 44 families (and an unclassified PL0 family) have been hierarchically classified in the CAZy database (<http://www.cazy.org/> [\[11\]](#)), and at least one member of each family has been subjected to activity identification and biochemical property analysis (Drula, et al., 2022 [\[12\]](#)). The identified GAG lyases are widely distributed in the PL6, PL8, PL12, PL13, PL15, PL16, PL21, PL23, PL29, PL30, PL33, PL35 and PL37 families (Drula, et al., 2022 [\[12\]](#)). Based on substrate specificity, GAG lyases are classified as HA-specific lyases, which can degrade HA only, CS/DS lyases (chondroitinase, CSases), which degrade CS/DS as well as HA in most cases, and Hep/HS lyases (heparinases, Hepases), which specifically cleave Hep/HS. All identified GAG lyases have substrate specificity to strictly distinguish one or two kinds of GAGs

based on saccharide composition and glycosidic bonds between each HexUA-HexNAC disaccharide units, while other uronic acid-containing polysaccharides are usually unable to serve as substrates for these enzymes.

With the evolution of host polysaccharides (Csoka and Stern, 2013; Popper, et al., 2011), the bacterial enzymes that degrade these polysaccharides have also evolved to varying degrees in response to complex environments. As a kind of unbranched anionic extracellular polysaccharide produced by lower organisms algae and bacteria (such as brown alga (Popper, et al., 2011) and bacteria belonging to *Pseudomonas* (Evans and Linker, 1973) and *Azotobacter* (Clementi, 1997)), alginate containing no hexosamine and sulfation possesses a simpler structure than GAGs in animals and likely emerge earlier than GAGs during the course of evolution. Like GAGs, alginate is a linear polyanionic polysaccharide that contains HexUA residues; however, alginate possesses a completely different chemical structure, which is composed of D-mannuronate (M) and its C5 epimer L-guluronate (G), and M and G residues often alternate randomly to form heteropolyuronic blocks, including polyM composed of M residues linked by β 1-4 bonds, polyG composed of G residues linked by α 1-4 bonds, and polyMG/GM composed of alternating M and G linked by β/α 1-4 bonds (Pawar and Edgar, 2012). Likewise, a series of alginate lyases essential for the metabolism of alginate were identified from microorganisms and algae as well as lower marine animals, which belong to the PL5, PL6, PL7, PL8, PL14, PL15, PL17, PL18, PL31, PL32, PL34, PL36, PL38, PL39, PL41 and PL44 families (Drula, et al., 2022). Based on their specificity, these lyases are also divided into the following categories: M block-specific lyases (Itoh, et al., 2019; Zhu, et al., 2015), G block-specific lyases (Matsubara, et al., 1998; Yamasaki, et al., 2005) and MG-specific alginate lyases (Jagtap, et al., 2014; Yamasaki, et al., 2004). From an evolutionary perspective in previous reports, GAG lyases are thought to have originated from alginate lyases by a divergent evolution through adaptation to the evolution of substrate polysaccharides. Some similarities in the sequences and folds of these two types of enzymes support this idea. For example, the $(\alpha/\alpha)_n$ toroid and antiparallel β -sheet folds of GAG lyases are found in families such as PL8 (hyaluronate lyase and chondroitin lyase), PL12 (heparinase III), PL21 (heparinase II), and PL23 (chondroitin lyase). Alginate lyases in PL15, PL17, and PL39 families also exhibit these folds. Additionally, the β -jelly roll fold of heparinase I in the PL13 family and alginate lyases in PL7 and PL18 families show similar structures. Lastly, the β -helix fold of chondroitinase B and alginate lyase in the PL6 family is another example (Garron and Cygler, 2014). Considering that the alginate with simpler structure from brown alga or bacteria might appear earlier, it is possible that the ancestral enzymes originally used alginate as a substrate. However, enzymes with transitional characteristics in activity and structure have not been discovered to support this speculation.

Recently, a new class of GAG lyases belonging to PL35 family was discovered with an ultrabroad substrate spectrum (Wei, et al., 2024). Unlike the identified GAG lyases (HA-specific lyases, CSases and Hepases) with quite strict substrate specificities, these novel GAG lyases can degrade three types of uronic acid-containing GAGs (HA, CS and HS), and thus were named GAGases. More interestingly, one of the eight identified GAGases (GAGase III) can even degrade alginate. Analysis of substrate structure preference represented by GAGase I showed that GAGases selectively act on GAG domains composed of non/6-*O*-*N*-sulfated hexosamines and D-glucuronic acid, among which GAGase III can also selectively degrade polyM blocks composed of D-mannuronic acids but not polyG blocks composed of L-guluronic acids in alginate. Furthermore, the primary structure alignment and phylogenetic analysis showed that GAGases have considerable degree of similarity with not only GAG lyases from PL15, PL21 and PL33 families but also alginate lyases from PL15, PL17 and PL34 families (Wei, et al., 2024), indicating that GAGases may exhibit some transitional features from alginate lyases to GAG lyases in structure.

In this study, crystal structures of two GAGases (GAGase II and GAGase VII) were determined for the first time, and their catalytic mechanism was further elucidated through the site-directed mutagenesis of their crucial site residues and molecular docking. Structural alignment showed

that GAGases are more structural similarity to some alginate lyases rather than GAG lyases, suggesting that PL35 family GAGases might originate from alginate lyases possessing an N-terminal $(\alpha/\alpha)_n$ toroid domain and a C-terminal antiparallel β -sheet domain. Moreover, one of the reasons why GAGase III can additionally degrade M-block alginate are also resolved by structural modeling, structural alignment and site-directed mutagenesis. This study not only helps to resolve the catalytic mechanism of the PL35 family lyases in particular GAGases, but also provides potential structural evidence for the divergent evolution of GAG lyases.

Results

Overall crystal structures of GAGase II and GAGase VII

To explore the substrate recognition and catalytic mechanism of the GAGases, the eight enzymes with different sequence identity (*Supplemental Table S1*) were individually used to prepare their crystals for structural analysis. The structures of GAGase II from *Spirosoma fluviale* (GenBank accession number: SOD82962.1, 81.2% sequence identity with GAGase I), GAGase VII from *Bacteroides intestinalis* (GenBank accession number: EDV05210.1, 44.8% sequence identity with GAGase I) and selenomethionine (SeMet)-labelled GAGase II (Se- GAGase II) in their ligand-free states were solved at 1.9 Å, 2.4 Å and 2.0 Å resolution, respectively (**Table 1**). Both GAGase II and GAGase VII are monomers consisting of two domains: a N-terminal $(\alpha/\alpha)_6$ toroid domain and a C-terminal two-layered β -sheet domain (**Figure 1A**).

The N-terminal domain (Met³⁴-Ala³⁶⁵) of GAGase II is composed of 16 large or small α -helices ($\alpha 1$ - $\alpha 16$) to form an $(\alpha/\alpha)_6$ toroid domain. Specifically, the N-terminal 16 α -helices constitute seven complete α -helix pairs ($\alpha 1$ -2, $\alpha 3$ -4, $\alpha 5$ -7, $\alpha 8$ -9, $\alpha 10$ -12, $\alpha 13$ -14 and $\alpha 15$ -16), forming seven hairpin structures, which eventually embrace into the $(\alpha/\alpha)_6$ toroid structure. The transition between $\alpha 6$ to $\alpha 8$ and between $\alpha 10$ to 12 is discontinuous, and both transitions possess a small α -helix $\alpha 7$ and $\alpha 11$, respectively. In each hairpin structure, two antiparallel α -helices are connected by loops composed of 9-12 residues, and each hairpin contains a turn structure composed of 1-4 residues. The N-terminal of this toroid domain is connected to a signal peptide that has been removed during cloning expression, and its C-terminal is connected to the two-layered β -sheet domain through a 17-residue linkage (Trp³⁵⁰-Met³⁶⁵). Its inner layer is formed by helices $\alpha 1$, $\alpha 3$, $\alpha 5$, $\alpha 8$, $\alpha 10$, $\alpha 13$ and $\alpha 16$, while its outer layer is formed by $\alpha 2$, $\alpha 4$, $\alpha 6$ + $\alpha 7$, $\alpha 9$, $\alpha 11$ + $\alpha 12$, $\alpha 14$ and $\alpha 15$. Moreover, the α -helix pairs $\alpha 13$ -14 and $\alpha 15$ -16 are significantly shorter than the first five sets, and $\alpha 15$ and $\alpha 16$ are connected by a short β -turn consisting of only 3 residues, which makes the toroid structure closed (**Figure 1A**).

The C-terminal domain (Met³⁶⁶-Ser⁶¹²) of GAGase II is composed of 17 β -strands ($\beta 1$ - $\beta 17$) to form a two-layered β -sheet domain with 4 small α -helices located between $\beta 5$ - $\beta 6$ and $\beta 8$ - $\beta 9$. Both layers of this C-terminal β -sheet domain are composed of eight groups of β -strands, namely $\beta 1$ - $\beta 5$, $\beta 11$ and $\beta 15$ - $\beta 16$ and $\beta 7$ - $\beta 10$, $\beta 12$ - $\beta 14$ and $\beta 17$. The two layers are roughly parallel to each other and twisted to form an angle of approximately 60°. The first five β -strands ($\beta 1$ - $\beta 5$) of the C-terminal domain are arranged in anti-parallel configuration. Subsequently, there is an Ω -loop structure wrapping around a metal ion (Mn²⁺), which consists of three small α -helices and a small β -strands, with many residues (such as His⁴⁰¹, His⁴⁰³, Tyr⁴²⁷, Glu⁴³¹, Trp⁴³⁸, Arg⁴⁴⁶ and so on) on top, which might serve as potential substrate-binding sites. Finally, the C-terminal domain terminates at $\beta 17$ in the second layer of the bilayer structure. Taken together, these structural elements constitute the C-terminal anti-parallel bilayer β -sheet domain of GAGase II (**Figure. 1A**).

Similar to GAGase II, GAGase VII also has an N-terminal $(\alpha/\alpha)_6$ toroid domain (Leu³⁵-Arg³⁶⁰) composed of 16 α -helices, but its $\alpha 1$ helix consists of only three residues (Glu⁴²-Val⁴⁴) and a long loop chain (**Figure. 1A**). Moreover, the C-terminal two-layered β -sheet domain (Met³⁶¹-Met⁶¹⁶) of GAGase VII is also composed of 17 β -strands ($\beta 1$ - $\beta 15$, $\beta 16a$, $\beta 16b$ and $\beta 17$). However, unlike

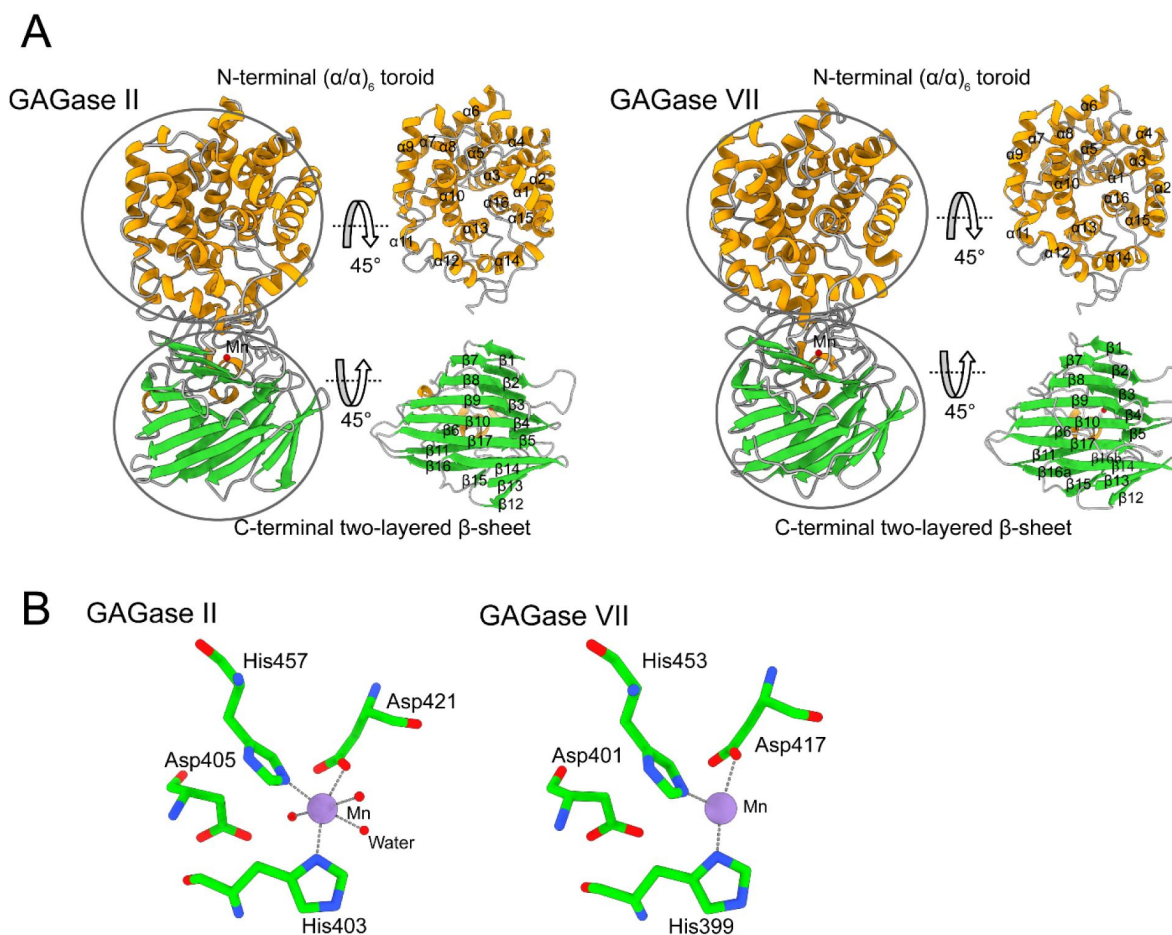


Figure 1.

Structural description of GAGase II and GAGase VII.

(A) Overall structures of GAGase II (*left*) and GAGase VII (*right*) are shown in cartoon. The α -helix, β -strand and random coil are colored with yellow, green and gray, respectively. The N-terminal $(\alpha/\alpha)_6$ toroid domain and C-terminal two-layered β -sheet domain were circled and the secondary structure elements are marked nearby. **(B)** Mn binding site of GAGase II and GAGase VII. The purple sphere presents Mn^{2+} . Mn^{2+} binding site of GAGase II (*left*) and GAGase VII (*right*) is shown in stick. A cut off distance of 3.0 Å was carried out to choose neighboring residues.

Table 1.

Data collection and refinement statistics.

	GAGase II	GAGase VII	SeMet-GAGase II
PDB entry code	8KHV	8KHW	-
Data collection			
Wavelength (Å)	0.98	0.98	0.98
Resolution range	44.01-1.90 (1.94-1.90)	19.87-2.40 (2.49-2.40)	40.91-2.00 (2.07-2.00)
Space group	$P2_1$	$P4_32_12$	$P2_1$
Unit cell	a=45.92 Å b=82.10 Å c=79.89 Å $\alpha=90^\circ$ $\beta=106.59^\circ$ $\gamma=90^\circ$	a=98.41 Å b=98.41 Å c=134.75 Å $\alpha=90^\circ$ $\beta=90^\circ$ $\gamma=90^\circ$	a=46.39 Å b=81.82 Å c=79.66 Å $\alpha=90^\circ$ $\beta=106.88^\circ$ $\gamma=90^\circ$
Unique reflections	86421 (4092)	49305 (2737)	38212 (3738)
Completeness (%)	98.55 (92.04)	99.68 (99.85)	99.15 (98.47)
R_{meas}	0.10 (0.23)	0.10 (0.74)	0.08 (0.24)
$R_{\text{p.i.m}}$	0.04 (0.10)	0.02 (0.16)	0.03 (0.093)
Mean $I/\sigma(I)$	12.9 (6.1)	20.6 (4.3)	17.9 (8.2)
$CC_{1/2}$	0.99 (0.96)	0.99 (0.97)	0.99 (0.97)
Refinement statistics			
R_{work}	0.16 (0.17)	0.2079 (0.26)	-
R_{free}	0.1919 (0.23)	0.2722 (0.36)	-
RMSD bond length (Å)	0.006	0.008	-
RMSD bond angle (°)	0.79	0.99	-
Ramachandran plot (%)			
Favored	97.75	94.31	-
Allowed	2.25	5.00	-
Outliers	0.00	0.69	-
Rotamer outliers (%)	0.00	0.40	-
Clashscore	2.78	6.89	-
Average B-factor (Å ²)	13.51	54.14	-
Macromolecules	11.66	54.20	-
Ligands	21.71	45.54	-
Solvent	25.75	47.24	-

Statistics for the highest-resolution shell are shown in parentheses.

GAGase II, its β 16 strands is interrupted into two small β -strands (β 16a, β 16b) by a randomly coiled loop chain. Compared with GAGase II, these additional loop regions may make GAGase VII less structurally stable than GAGase II, and may be an important factor in why the activity of GAGase VII is far lower than GAGase II. Nevertheless, structural alignment further shows that GAGase II has good superposition with GAGase VII and other GAGases, indicating that GAGases are highly conserved in structures (*Supplemental Figure S1*).

Mn binding site of GAGase II and GAGase VII

Unlike the GAG lyases identified from PL8, PL15 and PL21 families, which contain Ca^{2+} or Zn^{2+} in their overall crystal structures (Shaya, et al., 2008 [\[1\]](#); Shaya, et al., 2006 [\[2\]](#); Zhang, et al., 2021 [\[3\]](#)), a Mn^{2+} was detected in C-terminal domains of both GAGase II and VII by inductively coupled plasma-mass spectrometry (ICP-MS) analysis (**Figure 1B** [\[4\]](#), *Supplemental Table S1*). Structurally, the Mn^{2+} is located inside the long Ω loop after the β 1- β 5 antiparallel β -sheet, and coordinates with the nearby nitrogen and oxygen atoms of several residues such as Asp and His in both GAGase VII and GAGase II (GAGase II also has coordination with three surrounding water molecules), with distances of 2.0-2.5 Å. Such interaction should stabilize the nearby loop structures and may form a substrate-binding site relevant for substrate selectivity. To investigate the roles of these adjacent residues interacting with the Mn^{2+} , His⁴⁰³, Asp⁴⁰⁵, Asp⁴²¹ and His⁴⁵⁷ in GAGase II were individually mutated to alanine (Ala), and the results showed that these variants resulted in complete or substantial inactivation of GAGase II (**Figure 2A** [\[5\]](#)). In addition, the inhibitory effects of the chelator (EDTA) and the stimulation of Mn^{2+} on the enzymatic activities also support the critical role of Mn^{2+} on GAGases (**Figure 2B** [\[6\]](#)).

Multiple structural alignments of GAGase II and its structurally similar GAG lyases and alginate lyases

The ultrabroad substrate degradation spectra of GAGases indicate that they should be structurally similar with some other PLs with different substrate specificities. By using GAGase II as a representative, a structural alignment assay showed that the PL35 GAGases are structurally similar to various N-terminal (α/α)_n toroid domain and C-terminal antiparallel β -sheet domain-containing GAG/alginate lyases from PL8 (Fethiere, et al., 1999 [\[7\]](#); Huang, et al., 2003 [\[8\]](#); Li and Jedrzejewski, 2001 [\[9\]](#); Lunin, et al., 2004 [\[10\]](#); Shaya, et al., 2008 [\[1\]](#)), PL12 (Hashimoto, et al., 2014 [\[11\]](#); Ulaganathan, et al., 2017 [\[12\]](#)), PL15 (Ochiai, et al., 2010 [\[13\]](#); Zhang, et al., 2021 [\[3\]](#)), PL17 (Park, et al., 2014 [\[14\]](#)), PL21 (Shaya, et al., 2006 [\[2\]](#)), PL23 (Sugiura, et al., 2011 [\[15\]](#)) and PL39 (Ji, et al., 2019 [\[16\]](#)). Notably, GAGase II shares very low sequence identity with PL21 heparinase II from *Pedobacter heparinus* (Shaya, et al., 2006 [\[2\]](#)) (23.1% identity with 29% query coverage), PL12 heparinase III from *Bacteroides thetaiotaomicron* (Ulaganathan, et al., 2017 [\[12\]](#)) (23.6% identity with 32% query cover) and PL39 alginate lyase from *Defluviitalea phaphyphila* (Ji, et al., 2019 [\[16\]](#)) (26.5% identity with 31% coverage), and no significant sequence identity with PL17 alginate lyases (Park, et al., 2014 [\[14\]](#)), PL15 exoHep (Zhang, et al., 2021 [\[3\]](#)), PL8 chondroitin ABC lyase (Shaya, et al., 2008 [\[1\]](#)) and chondroitin AC lyase (Lunin, et al., 2004 [\[10\]](#)); however, all are relatively well superimposable in their overall structures and highly conserved in their triplet active site residues (**Figure 3** [\[17\]](#)). In terms of overall structural similarities, the (α/α)_n toroid domain and anti-parallel β -sheet domain of GAGase II show more structural similarity to alginate lyases with (α/α)₆ toroid and the antiparallel β -sheet domain, such as those from the PL15 (PDB code: 3A00) (Ochiai, et al., 2010 [\[13\]](#)), PL17 (PDB code: 4NEI) (Park, et al., 2014 [\[14\]](#)) and PL39 (PDB code: 6JP4) (Ji, et al., 2019 [\[16\]](#)) families, than various GAG lyases (**Table 2** [\[18\]](#),3), indicating that GAGases might have originated from alginate lyases rather than GAG lyases. Reasonable support for this speculation could be the significant alginate-degrading activity of GAGase III. Therefore, the broad substrate spectrum of GAGases may be because they are evolutionarily transitional types with the enzymatic features of alginate lyases and GAG lyases.

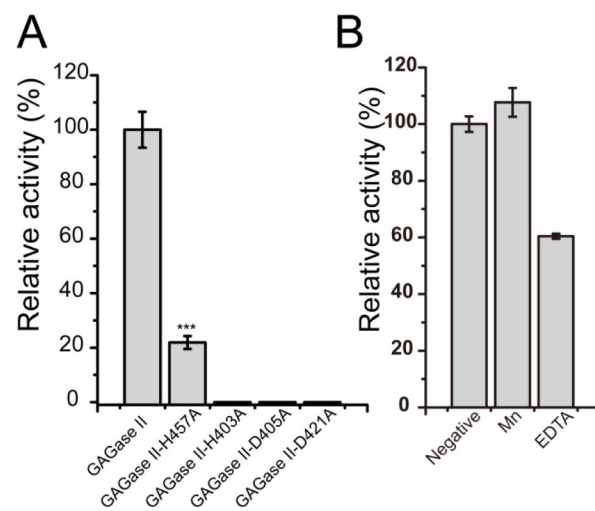


Figure 2.

Activity of GAGase II and its variants.

(A), Activity of site-directed mutants of His and Asp nearby the Mn^{2+} . His and Asp residues nearby the Mn^{2+} were individually mutated to Ala and the relative activity of each variant was measured as described in "Materials and methods". All of the residual activities were evaluated and shown as the relative intensity compared with that of wild type GAGase II. ***: $p < 0.001$. **(B)**, The effects of Mn^{2+} and chelating reagent of GAGase II were determined using HA (1 mg/ml) as substrate. Error bars represent averages of triplicates $\pm$ S.D.

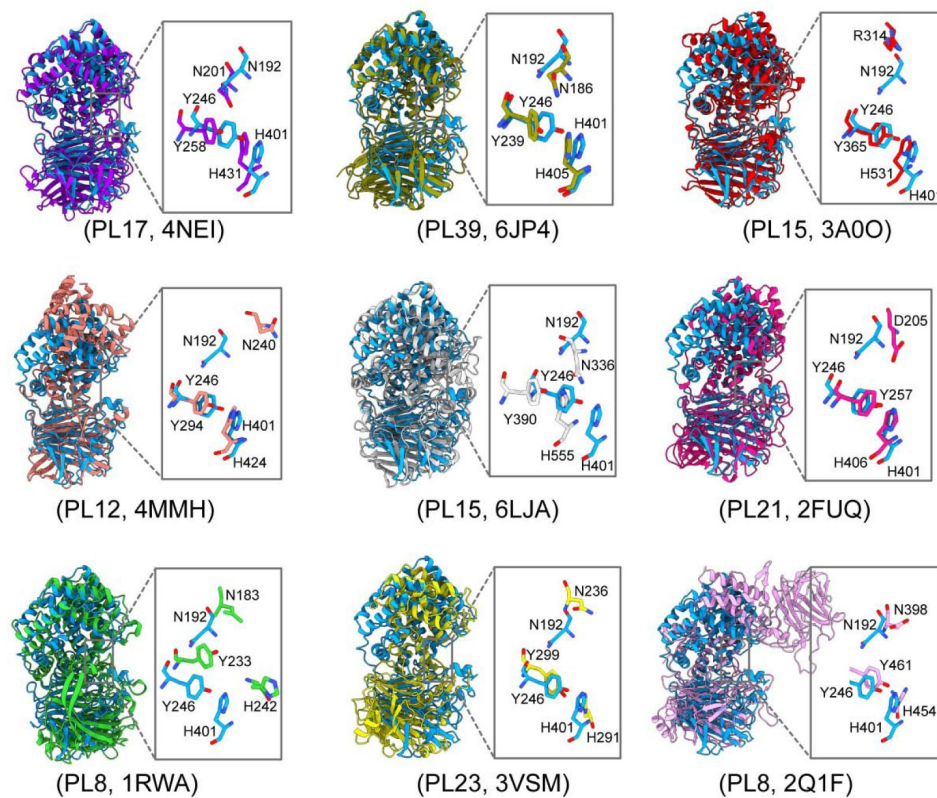


Figure 3.

Multiple structural alignments of GAGase II and its structurally similar proteins.

GAGase II (8KHV, *blue*) was aligned with structurally identified GAGs and alginate lyases, including PL17 family alginate lyase (4NEI, *purple*), PL39 family alginate lyase (6JP4, *olive*), PL15 family alginate lyase (3A0O, *red*), PL12 family heparinase III (4MMH, *salmon*), PL15 family exoHep (6LJA, *gray*), PL21 family heparinase II (2FUQ, *magenta*), PL8 family chondroitin sulfate AC lyase II (1RWA, *green*), PL23 family chondroitinase (3VSM, *yellow*) and PL8 family chondroitin sulfate ABC lyase II (2Q1F, *pink*).

Table 2.

Structural similarity of GAGase II/GAGase VII with GAG/alginate lyases analysed using DALI.

	Chain	Z-score	RMSD	lali	nres	Description	PL family
			(Å)				
GAGase II	3a0o-A	29.4	3.2	539	764	Alginate lyase	15
	6jp4-A	28.9	3.0	523	770	Alginate lyase	39
	4nei-A	27.3	3.1	523	705	Alginate lyase	17
	4mmh-A	28.6	4.4	481	637	Heparinase III	12
	6lja-A	25.7	3.9	540	841	Exo-type heparinase	15
	2fuq-A	25.1	4.9	528	743	Heparinase II	21
	1rwa-A	15.2	5.4	470	754	Chondroitin AC lyase	8
	3vsm-a	14.3	5.3	369	633	Chondroitinase	23
	2q1f-a	11.1	5.3	434	991	Chondroitin ABC lyase	8
GAGase VII	3a0o-A	29.2	3.6	545	764	Alginate lyase	15
	6jp4-A	28.0	3.3	530	770	Alginate lyase	39
	4nei-A	27.2	3.2	529	705	Alginate lyase	17
	4mmh-A	26.4	4.4	492	637	Heparinase III	12
	2fuq-A	23.1	5.4	531	747	Heparinase II	21
	1rwa-A	14.3	4.5	473	754	Chondroitin AC lyase	8
	3vsm	12.9	4.3	365	633	Chondroitinase	23
	2q1f-A	10.5	4.6	407	991	Chondroitin AC lyase	8

Table 3.

Multiple structural alignments of $(\alpha/\alpha)_n$ toroid domain or antiparallel β -sheet domain of GAGase II and identified GAG/alginate lyases

		PL family	Domain	Aligned atoms	RMSD	Substrate
$(\alpha/\alpha)_n$ toroid domain	8KHV	PL35	$(\alpha/\alpha)_6$ toroid (Met ²⁴ -Trp ³⁵⁰)	317	0.00	HA, CS and HS
	3A0O	PL15	$(\alpha/\alpha)_6$ toroid (Gly ¹²⁷ -Leu ⁴⁵²)	101	1.04	Alginate
	4NEI	PL17	$(\alpha/\alpha)_6$ toroid (His ³⁰ -Glu ³⁶⁴)	102	1.20	Alginate
	6JP4	PL39	$(\alpha/\alpha)_6$ toroid (Ala ¹ -Tyr ³⁵⁵)	113	1.30	Alginate
	2FUQ	PL21	$(\alpha/\alpha)_6$ toroid (Thr ²⁷ -Leu ³⁶⁸)	67	1.42	Hep, HS
	1RWA	PL8	$(\alpha/\alpha)_6$ toroid (Pro ⁴ -Val ³⁶⁷)	60	1.24	HA, CS
	3VSM	PL23	$(\alpha/\alpha)_5$ toroid (Asn ⁷² -Asn ³³⁰)	22	1.14	CS
	7FHU	PL5	$(\alpha/\alpha)_6$ barrel (Cys ²³ -Pro ³²⁹)	27	1.105	Alginate
	8BDQ	PL38	$(\alpha/\alpha)_7$ barrel (Ala ²⁴ -Lys ⁴⁰²)	58	1.177	Alginate
	4MMH	PL12	$(\alpha/\alpha)_5$ toroid (Ile ²⁹ -Thr ³⁷⁹)	24	1.169	Hep, HS
antiparallel β -sheet domain	8KHV	PL35	anti-parallel β -sheet	261	0.00	HA, CS and HS
	3A0O	PL15	anti-parallel β -sheet	109	1.11	Alginate
	4NEI	PL17	anti-parallel β -sheet	85	0.97	Alginate
	6JP4	PL39	anti-parallel β -sheet	67	1.06	Alginate
	2FUQ	PL21	anti-parallel β -sheet	103	1.17	Hep, HS
	4MMH	PL12	anti-parallel β -sheet	97	1.22	Hep, HS

The detailed views of the crucial active site residues are shown in stick mode. The root-mean-square deviation (RMSD) between these structures and GAGase II were calculated as 1.34, 1.34, 1.30, 1.24, 1.15, 1.35, 1.24, 1.14 and 1.41 Å based on 171, 148, 141, 111, 99, 87, 60, 22 and 21 pruned atoms, respectively.

Catalytic center and substrate binding sites of GAGases

Based on the sequence and structural alignment with homogenous alginate/GAG lyases, the conserved residues Tyr²⁴⁶, His⁴⁰¹, Asn¹⁹² of GAGase II, Tyr²⁴³, His³⁹⁸, Asn¹⁸⁹ of GAGase III, and Tyr²⁴¹, His³⁹⁷, Asn¹⁸⁷ of GAGase VII may be the key triplet residues involved in β-elimination catalysis (**Figure 3**). To confirm this hypothesis, these residues were individually replaced by Ala and the activity of the mutants was analyzed. The results showed that the activity of all variants toward CS-C was completely lost (**Figure 4A**), indicating that GAGases act through a general acid base catalytic mechanism by using His/Tyr as a Brønsted acid/base, similar to that observed for various other identified GAG/alginate lyases (**Table 4**). Compared with structurally similar GAG/alginate lyases, the results from molecular docking and catalytic tunnel predictions showed that GAGase II contains a shorter catalytic cavity (*Supplemental Figure S2*), which may explain why they can accommodate and degrade a variety of substrates with very different structures. However, we were unsuccessful in preparing the cocrystals of GAGases/their inactive variants with substrates to reveal their substrate binding and degradation mechanisms. Alternatively, molecular docking experiments of GAGase II with several structurally defined hexasaccharide (Hexa) substrates were performed as shown in **figure 4B**. Based on the docking results obtained for GAGase II with an HA Hexa (GlcUA1–3GlcNAc)₃ (PDB code: 1HYA) (Winter, et al., 1975) (**Figure 4B**), the nonreducing end of the substrate at the “-1” and “-2” subsites is near some positively charged amino acids, including Arg⁸⁷, Arg⁸⁸, Arg⁹⁴, His¹³⁶, Asn¹⁹¹ and Asn¹⁹², in the vicinity of the catalytic cavity of GAGase II, which could facilitate the binding of negatively charged substrates. Besides, several acidic amino acids (Glu²⁴², Asp²⁹⁹ and Glu⁴³¹) are very close to the “+1” and “+2” subsites, which could promote the release of the reducing end product via charge repulsion. There are some other residues surrounding around the catalytic cavity, such as Tyr²⁴⁰, Gln³⁰⁷, Arg³⁴², Tyr⁴²⁷, Glu⁴²⁸, Glu⁴³¹, Trp⁴³⁸, Met⁴⁴⁰ and Arg⁴⁴⁶, which may play a key role in stabilization of the catalytic cavity and overall structure. Site-directed mutagenesis of these residues individually replaced with Ala showed that resulting variants were partially or completely inactivated (**Figure 4C**), confirming that these residues play important roles in the binding and release of substrate and structural stabilization.

Substrate selectivity of GAGases

As mentioned above, GAGase III can degrade not only HA, CS and HS but also alginate, even though the activity on alginate is relatively low (about 1/50 of the activity on HA). To investigate the reason why this enzyme is capable to act on alginate, its structure was predicted using RoseTTAfold, and aligned with GAGase II and GAGase VII. Results showed that GAGase III has a unique His¹⁸⁸ residue in its catalytic cavity, which is conserved in alginate lyases from PL17 (Park, et al., 2014), PL39 (Ji, et al., 2019), PL38 (Ronne, et al., 2023), PL5 (Pandey, et al., 2021) and PL15 (Ochiai, et al., 2010) families, while other GAGases have a conserved asparagine residue such as the Asn¹⁹¹ of GAGase II and the Asn¹⁸⁶ of GAGase VII in the corresponding position (**Figure 5A-B**). To verify the function of this histidine, His¹⁸⁸ was replaced with alanine and asparagine, respectively. The results of substrate-specificity analysis showed that the alginate-degrading activity of both GAGase III-H188A and GAGase III-H188N were abolished even at a quite high ratio of the mutated enzyme to substrate such as 30 μg enzyme to 30 μg substrate (*Supplemental Figure S3A*), while their GAG-degrading activity was only partially affected, indicating that the His¹⁸⁸ residue is essential for the alginate-degrading activity of GAGase III (**Figure 5C-D**). However, when the asparagine residue at the corresponding position of other

Table 4.

Catalytic residues comparison of identified GAGs and alginate lyases shared similar structure with GAGase II and GAGase VII

	PDB code	PL family	Substrates	Sequence identity (Query cover/Per. Ident) *	Catalytic residues	
					General base/acid	Neutralizer
GAGase II	8KHV	PL35	HA, CS and HS	100%/100%	Tyr246 His401	Asn192
GAGase VII	8KHW	PL35	HA, CS and HS	93%/44.67%	Tyr241, His397	Asn187
GAGase III	-	PL35	HA, CS, HS and alginate (M-specific)	98%/64.08%	Tyr243,His398,	Asn189, His188
Chondroitin sulfate AC lyase II	1RWA	PL8	HA and CS	N.D.**	His233 Tyr242,	Asn183
Chondroitin sulfate ABC lyase II	2Q1F	PL8	HA, CS and DS	N.D.	His345,Tyr461,His454	Asp398
Heparinase III	4MMH	PL12	HS	8%/30.91%	Tyr294, His424	Asn240
Alginate lyase	3A0O	PL15	Alginate (M/G specific)	43%/23.93%	His311,Tyr365,His531	Arg314
Exo-Heparinase	6LJA	PL15	Hep and HS	N.D.	His337,Tyr390,His555	Asn336
Alginate lyase	4NEI	PL17	Alginate (M- specific)	15%/25.51%	Tyr258, His413	Asn201, His200
Heparinase II	2FUQ	PL21	Hep and HS	29%/23.13%	His202, Tyr257, His406	Glu205
Chondroitin lyase	3VSM	PL23	CS	N.D.	Tyr299,His291	Asn236
Alginate lyase	6JP4	PL39	Alginate (M/G specific)	31%/26.53%	Tyr239, His405	Asn186,His187

* Sequence identity means the sequence similarity compared with GAGase II.

**N.D. means no significant identity is detected.

GAGases was mutated to histidine, the obtained variants did not exhibit any alginate-degrading activity (*Supplemental Figure S3B*), suggesting that degrading activity of GAGase II remains to be determined outside of the His¹⁸⁸ residue.

GAGases show preference for some specific sulfation patterns (Wei, et al., 2024 [\[4\]](#)). For example, GAGases prefer to degrade CS domains composed of non-6-*O*-sulfated but not 4-*O*-sulfated GalNAc residues and D-GlcA residues. To further investigate the reason why these enzymes have such structural preference for substrates, we tried to prepare the co-crystal of GAGase II or VII with various structure-defined oligosaccharide substrates, but ultimately failed. Therefore, we used molecular docking analysis to preliminarily explore the possible reasons why GAGases have substrate structural preference. To this end, docking of GAGase II with a nonsulfated CS Hexa (GlcUA β 1–3GalNAc)₃ (PDB code: 2KQO) and a tri-4-*O*-sulfated CS Hexa (GlcUA β 1–3GalNAc(4S))₃ (PDB code: 1C4S) (Sattelle, et al., 2010 [\[4\]](#)) were carried out, respectively. In the docked model, the monosaccharide residues of the substrate were named “+1” and “+2” toward reducing end, and “-1” and “-2” toward non-reducing end from the β 1-4 cleavage site following the nomenclature (Davies, et al., 1997 [\[4\]](#)). The results show that for the nonsulfated CS Hexa, the GlcUA residue at the “+1” subsite is closely approached by the Tyr²⁴⁶ residue as a Brønsted base (**Figure 6 left** [\[4\]](#)), which facilitates the uptake of the C5 proton of GlcUA by Tyr²⁴⁶, and in the case of tri-4-*O*-sulfated CS Hexa, the 4-*O*-sulfate group in GalNAc residue at the “+2” subsite interacts with the neutralizer Asn¹⁹² and the catalytic residue Tyr²⁴⁶ via hydrogen bonds etc., the C5 proton of GlcUA at the “+1” subsite is away from the key catalytic residue Tyr²⁴⁶ (**Figure 6 right** [\[4\]](#), *Supplemental Figure S4*), which hinders the degradation of CS with such sulfation pattern, as we detected in the biochemical analysis above. Of course, the results from docking assay need to be further confirmed by more structural and biochemical evidence.

Catalytic mechanism of GAGases

In summary, taking GAGase II as representative, the catalytic process of GAGases is proposed as follows. The negatively charged GAG substrate binds to several positively charged residues, such as Arg⁸⁷, Arg⁸⁸, Arg⁹⁴, His¹³⁶, Asn¹⁹¹ and Asn¹⁹², near the catalytic triad residues in the catalytic cavity. The Asn¹⁹¹ and Asn¹⁹² residues interact with the carboxyl group of GlcUA at the +1 subsite to reduce the pKa and promote the protonation of the C5 proton. The His⁴⁰¹ works as a proton receptor (Brønsted base) to abstract protons from the C5 position of GlcUA and create an enolate tautomeric. The protonated Tyr²⁴⁶ (Brønsted acid) provides a proton to the C-4–O-1 glycosidic bond between the “-1” and “+1” subsite. The glycosidic bond is cleaved as an electron transfer occurs from the carboxyl group to form a double bond between C4 and C5 of GlcUA. A new reducing end and a new unsaturated nonreducing end are recreated. Moreover, the degradation of polyM by GAGase III requires the involvement of His¹⁸⁸ residue, which may involve in the neutralization of carboxyl groups in M-blocks in alginate (**Figure 7** [\[4\]](#)).

Discussion

GAG lyases derived from microorganisms, as GAG-specific degrading enzymes, are classified into 13 PL families in the CAZy database (Drula, et al., 2022 [\[4\]](#)). They are not only critical for microorganisms to degrade and utilize GAGs as carbon sources, but also become useful tools for structural and functional studies of GAGs. In contrast, alginate lyases, as alginate-specific degrading enzymes, are classified into 15 PL families in the CAZy database (Drula, et al., 2022 [\[4\]](#)). They are essential for the metabolism of alginate in microorganisms and algae as well as lower marine animals, and have completely different substrate specificities than GAG lyases. Due to the similarity between GAG lyases and alginate lyases in 3D fold, GAG lyases are thought to have originated from the divergent evolution of alginate lyases (Garron and Cygler, 2014 [\[4\]](#)), but enzymes that exhibit transitional features in function and structure, which would provide direct

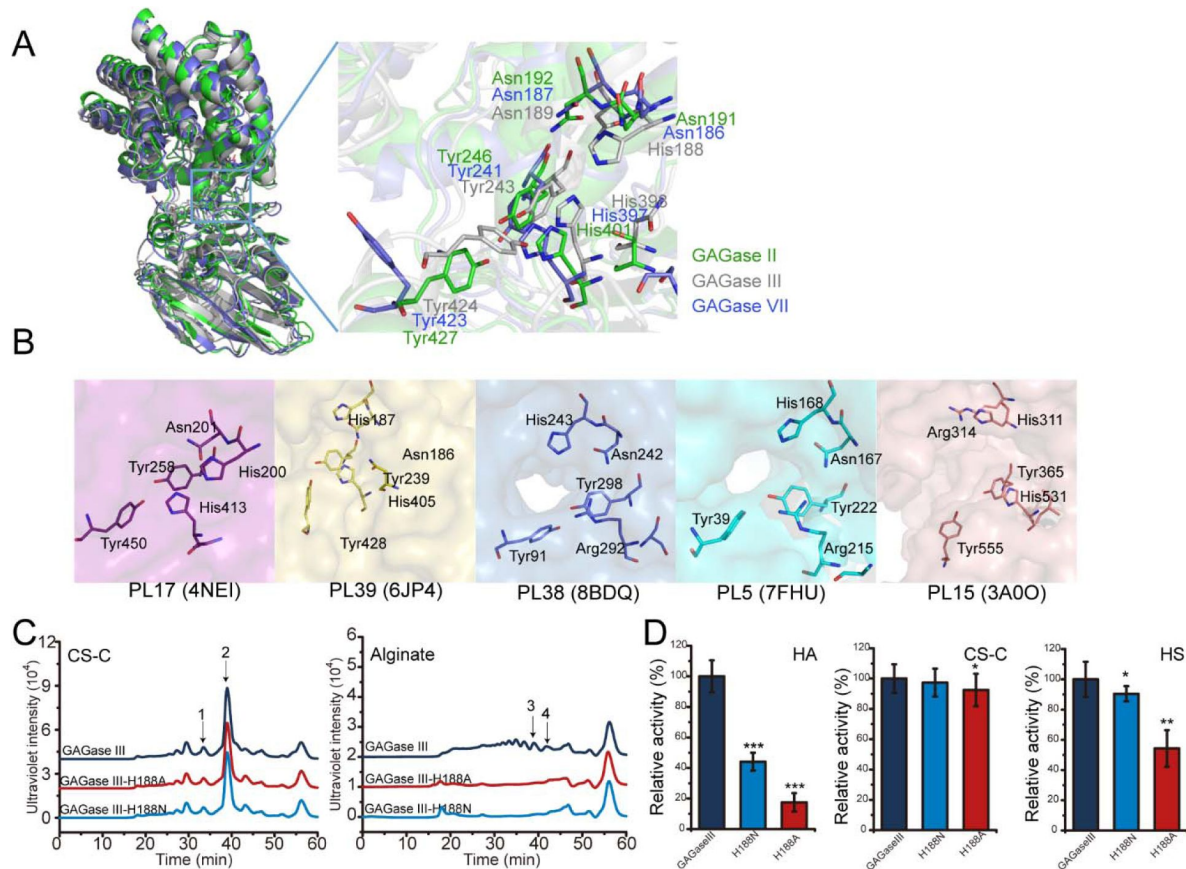


Figure 5.

Analysis of a key residue for the alginate-degrading activity of GAGase III

(A), Multiple structural alignment of GAGase II, III and VII. GAGase II (8KHV, *green*) and GAGase VII (8KHW, *skyblue*) were aligned with GAGase III (model, *gray*). The detailed views of crucial catalytic residues are shown in stick mode; (B), Conserved catalytic residues of alginate lyases from PL17 (4NEI, *purple*), PL39 (6JP4, *yellow*), PL38 (8BDQ, *slate*), PL5 (7FHU, *cyan*) and PL15 (3A00, *pink*) family. Residues are shown in stick; (C), Activity assay of GAGase III-H188N and GAGase III-H188A toward CS-C and alginate. The crucial site His¹⁸⁸ was mutated to alanine and asparagine, respectively. The activity of each variant against CS-C and alginate was assessed using gel filtration HPLC on a Superdex Peptide column, as described under “Materials and methods”; the elution of each fraction is indicated as follows: 1, CS-C tetrasaccharide; 2, CS-C disaccharide; 3, alginate trisaccharide; 4, alginate disaccharide; (D), Relative activity of GAGase III and its variants. Three types of GAGs (HA, CS-C and HS) were individually treated with GAGase III and its variants (GAGase III-H188N and GAGase III-H188A) at 40 °C for 1 h; relative activities of enzymes were determined by detecting the absorbance at 232 nm. Data are shown as the percentage of the activity relative to the wild-type GAGase III. *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$.

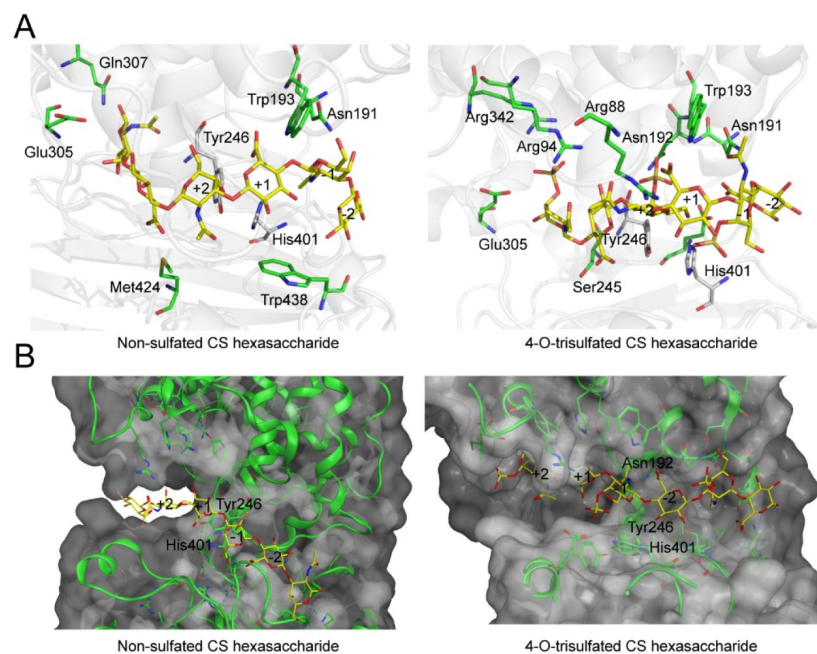


Figure 6.

Molecular docking of GAGase II with hexasaccharide ligands.

Molecular docking of GAGase II with CS ligands. The molecule docking was carried out with GAGase II and CS ligands, including nonsulfated (PDB code: 2KQO) (*left*) and 4-O-trisulfated (PDB code: 1C4S) (*right*) CS hexasaccharide, to investigate the substrate selectivity, using AutoDock Vina (A) and Molecular Operating Environment (MOE) (B). The catalytic triplet residues (*green*) and CS ligand (*yellow*) are showed as sticks.

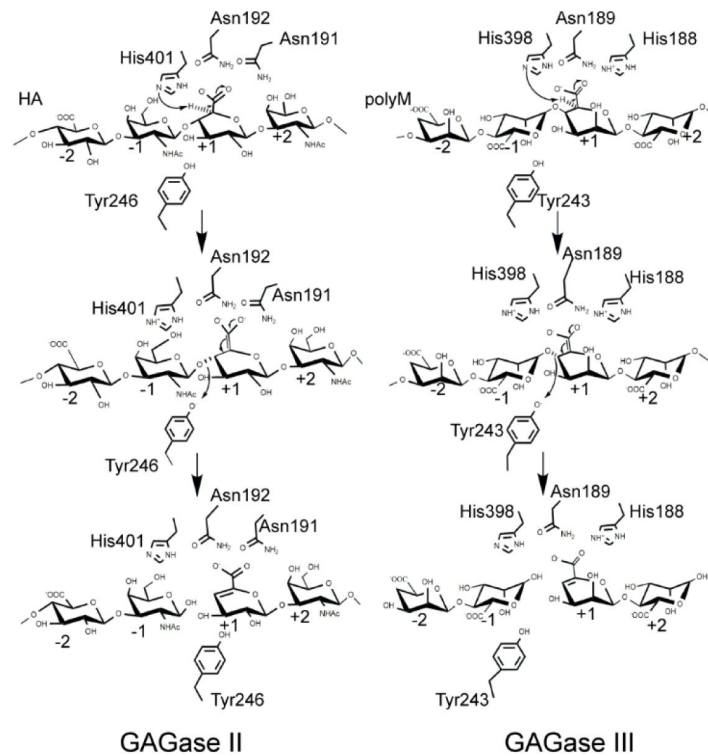


Figure 7.

Proposed catalytic mechanism of GAGase II and GAGase III.

Take the HA degradation by GAGase II and the polyM degradation by GAGase III for example. Briefly, the substrate is firstly binding to the negatively charged residues near the catalytic sites, the carboxylate group is neutralized by Asn¹⁹², Asn¹⁹¹ in GAGase II and His¹⁸⁸, Asn¹⁸⁹ in GAGase III; His³⁹⁸ in GAGase II and His⁴⁰¹ in GAGase III are proposed to work as general base to abstract a proton from the C5 position of GlcUA at +1 position, and Tyr²⁴⁶ in GAGase II and Tyr²⁴³ in GAGase III work as general acid to donate the leaving group at -1 position a proton. The arrows indicate the direction of electron transfer, and the C5-C6 double bond on the middle panel indicate the enolate anion intermediate created by proton abstraction at C-5 position.

evidence to support this view, are lacking. On the basis of previous functional studies (Wei, et al., 2024 [DOI](#)), structural studies of GAGases with HA, CS, HS and even alginate-degrading activities should further provide substantial evidence for the evolution of GAG lyases.

Prior to this study, no member of the PL35 family had been structurally characterized. Here, the structures of GAGase II and VII were determined at resolutions of 1.9 Å and 2.4 Å, respectively, and both exhibit a highly similar structure of “N-terminal (α/α)₆ toroid and C-terminal two-layer antiparallel β -sheet”, which is also the fold adopted by most structurally known GAG lyases except CSase B in PL6 (Huang, et al., 1999 [DOI](#)), Hepase I in PL13 (Han, et al., 2009 [DOI](#)) and hyaluronan lyase in PL16 (Martinez-Fleites, et al., 2009 [DOI](#)). Structural alignment has shown that GAGases share considerable degrees of similarity with all GAG/alginate lyases with the (α/α)_n toroid and antiparallel β -sheet fold in PL8 (Fethiere, et al., 1999 [DOI](#)), PL12 (Hashimoto, et al., 2014 [DOI](#)), PL15 (Ochiai, et al., 2010 [DOI](#); Zhang, et al., 2021 [DOI](#)), PL17 (Park, et al., 2014 [DOI](#)), PL21 (Shaya, et al., 2006 [DOI](#)), PL23 (Sugiura, et al., 2011 [DOI](#)) and PL39 (Ji, et al., 2019 [DOI](#)) families, especially the key catalytic triplet residues (Tyr, His and Asn) in their active centers, which are highly conserved; thus, they may share a similar Brønsted base/acid catalytic mechanism (Garron and Cygler, 2010 [DOI](#); Garron and Cygler, 2014 [DOI](#)). Notably, GAGases are more structurally similar to alginate lyases from PL15 (Ochiai, et al., 2010 [DOI](#)), PL17 (Park, et al., 2014 [DOI](#)) and PL39 (Ji, et al., 2019 [DOI](#)) family than to various GAG lyases, which further structurally supports the speculation that GAGases originated from alginate lyases.

As mentioned previously, many different GAG/alginate lyases share similar catalytic sites and the same catalytic machinery (Garron and Cygler, 2010 [DOI](#); Garron and Cygler, 2014 [DOI](#)), but their differences in substrate selectivity and endo-/exolytic manner are mainly attributed to subtle differences in the three-dimensional structure (Lombard, et al., 2010 [DOI](#)), which may be the substrate-binding sites. The recognition of appropriate substrates by PLs mainly depends on the interaction between the positively charged side chains of basic amino acid residues in their substrate-binding sites and the negatively charged carboxyl groups of HexUA residues in substrates. Moreover, compared to the structurally and biochemically characterized members of the PL8, PL12, PL15, PL21 and PL39 families, PL35 family proteins possess a shorter catalytic cavity; thus, PL35 family proteins may more easily accommodate various substrates with different structures but may exhibit a weaker ability to bind and degrade specific substrates, resulting in the activities of GAGases towards various GAGs and alginate are much lower than those of GAG or alginate-specific lyases. Certainly, the accurate substrate selectivity mechanism of GAGases remains to be revealed by preparing and resolving co-crystals of the enzyme- substrate complex in the future.

In addition, GAGases tend to act on GlcUA-containing HA, CS and HS but not IdoUA- containing DS and Hep, which should result from the absence of a His residue functioning as an *anti*-base in the catalytic cavity (Garron and Cygler, 2010 [DOI](#); Shaya, et al., 2008 [DOI](#); Shaya, et al., 2006 [DOI](#)). The His residue as an *anti*-base can substitute Tyr to absorb the C5 proton in the IdoUA and ensure that β -elimination is carried out successfully, which is detected in chondroitin sulfate ABC lyase (2Q1F) from PL8 family (His⁴⁵⁴) (Shaya, et al., 2008 [DOI](#)) and heparinase II (2FUQ) from PL21 family (His²⁰²) (Shaya, et al., 2006 [DOI](#)), and endows these enzymes with a bifunctional activity against GlcUA- and IdoUA-containing GAGs. Notably, GAGase III, the only identified GAGase with alginate-degrading activity, has a unique His¹⁸⁸ residue also conserved in various alginate lyases from PL5, PL15, PL17, PL38 and PL39 families. Mutation assay showed that the His¹⁸⁸ is a key residue for the alginate-degrading activity of GAGase III. However, when a conversed asparagine residue at the corresponding position of other GAGases was replaced by histidine, the muted GAGases still could not act on alginate, indicating that some other unidentified structural factors are also involved in the alginate degradation of GAGase III. Overall, while the substrate-degrading features of GAGases can be preliminarily elucidated structurally, more direct evidence remains to be obtained by detailed structural and biochemical approaches.

In addition, the identification of GAGases may help clarify an inappropriate naming issue regarding the functional module “Hepar_II_III superfamily”. This functional module is commonly found in not only GAG lyases but also alginate lyases with an “N-terminal (α/α)_n toroid + C-terminal β -sandwich” structure. Interestingly, however, a large number of identified alginate lyases with the functional module “Hepar_II_III superfamily” do not have the ability to degrade Hep/HS. Historically, due to the importance of Hep as an anticoagulant in clinical application, Hep/HS-related studies including heparinases have received much higher attention than alginate-related studies including alginate lyases, which may be the reason why some sequences in alginate lyases that are similar to heparinase are annotated named the “Hepar_II_III superfamily” module. From an evolutionary perspective, this “Hepar_II_III superfamily” module might be a conserved functional domain in the divergent evolution process from alginate lyase to GAG lyase. Therefore, it might be more reasonable and accurate for this functional domain to be named as the “alginate lyase superfamily” module, and the presence of this functional module in GAGases with evolutionary transitional characteristics may well support this view.

In conclusion, the first determination of the structure of GAGases not only facilitates the elucidation of the catalytic mechanism of PL35 family enzymes, especially the substrate selection mechanism of GAGases, but also provides potential evidence for the hypothesis that GAG lyase evolved from alginate lyase.

Materials and methods

Materials

HA from *Streptococcus equi*, CS-C from shark cartilage, alginate sodium from brown algae, polyethylene glycol (PEG) 6000, PEG 400, calcium chloride (CaCl_2), tris-(hydroxymethyl) aminomethane (Tris) and hydroxyethyl piperazine ethanesulfonic acid (HEPES) were purchased from Sigma Aldrich. Lysine, phenylalanine, threonine, leucine, isoleucine, valine and L-selenomethionine were purchased from Solarbio Co.,Ltd. (Beijing, China).

Heterologous expression and purification of GAGases and GAGase II selenomethionine derivant

The cloning and induced expression of PL35 family GAGases were described in previous study (Wei, et al., 2024²⁴). Briefly, the synthesized expression plasmid pET-30a-GAGases were transformed into *E. coli* BL21 (DE3) cells and culture in LB broth at 37 °C. Recombinant enzymes with a 6His-tag were expressed at 16 °C for 16 h by inducing with isopropyl 1-thio- β -D-galactopyranosid (IPTG) (final concentration of 0.05 mM) at the cell density of $\text{OD}_{600}=0.6-0.8$. Furthermore, to obtain the Se-GAGase II, cultured cells harboring the pET-30a-GAGase II were centrifuged at 3,700 g at 4 °C for 5 min, washed with M9 culture medium (17.2 g/L Na_2HPO_4 , 3 g/L KH_2PO_4 , 2.5 g/L NaCl and 5 g/L NH_4Cl), transferred into M9 culture medium supplemented with 1 mM MgSO_4 , 0.1 mM CaCl_2 , 0.4% (w/v) glucose and cultured at 37 °C until the OD_{600} reached 0.6. Then, the culture medium was cooled to 22 °C. Additional essential amino acids, including lysine (100 mg/L), phenylalanine (100 mg/L), threonine (100 mg/L), leucine (50 mg/L), isoleucine (50 mg/L), valine (50 mg/L), and L-selenomethionine (L-SeMet) (50 mg/L), were added and incubated at 22 °C for 15 min. The proteins contained L-SeMet were also induced and expressed at 16 °C for 16 h by supplementing with 5 mM IPTG.

After further induced cultivation, cells were harvested, resuspended and disrupted by sonication, the recombinant proteins in the supernatant of cell lysates are primarily purified by loading on a nickel affinity column, washing with buffer A containing 10 mM imidazole to remove impurities, and finally eluting using buffer A containing 250 mM imidazole. After nickel affinity chromatography, the elutes of GAGase II and GAGase VII were diluted and loaded on a Q-

Sepharose FF column (GE Healthcare) eluted with a gradient concentration of NaCl from 0 to 1 M, and the target proteins were desalted and concentrated through ultrafiltration and further sub-fractionated by gel filtration on a Superdex G-200 column (GE Healthcare) for crystal culture. The concentration of each protein was determined using BCA Protein Assay Kit with bovine serum albumin (BSA) as reference protein (Cwbio, Shanghai).

Activity assay of GAGase II, III, VII and their variants toward various GAGs and alginate

To evaluate the activity of GAGase II, III, VII and their variants, CS-C or alginate (30 μ g) was treated with each wild-type enzyme or its variant (6 μ g) in the optimal buffer (50 mM Tris-HCl, pH 7.0) for 12 h at 40 °C. In order to evaluate the activity of GAGase III-H188N and GAGase III-H188A in degrading alginate, alginate (30 μ g) was treated with each variant (3-30 μ g) at 40 °C for 12 h. After inactivated by boiling for 10 min, the products were analyzed by gel filtration HPLC using a SuperdexTM Peptide 10/300 GL column eluted with 0.20 M NH_4HCO_3 at a flow rate of 0.4 ml/min and monitored at 232 nm using a UV detector, and online analysis was conducted using LCsolution version 1.25.

To evaluate the activity of variants from the site-directed mutagenesis of some potential substrate-binding residues of GAGase II, HA (150 μ g) was treated with the wild-type or each mutant (6 μ g) in the optimal buffer (50 mM Tris-HCl, pH 7.0) at 40 °C for 12 h. To evaluate the activity of GAGase III-H188N and GAGase III-H188A, HA, CS-C, or HS (150 μ g) was treated with GAGase III and each variant (6 μ g) at 40 °C for 1 h. The absorbance of each resultant was determined at 232 nm using a UV spectrophotometer.

Crystallization, X-ray diffraction and data collection of GAGase II, GAGase VII and Se- Met-GAGase II

The purified GAGase II (20 mg/ml) was crystallized in 0.1 M HEPES (pH 7.0), 12% (w/v) PEG 6000 and 0.2 M CaCl_2 . The purified Se-Met-GAGase II (5 mg/ml) was crystallized in 0.1 M Tris (pH 7.5), 20% (w/v) PEG 6000 and 0.2 M CaCl_2 . The purified GAGase VII (10 mg/ml) was crystallized in 0.1 M HEPES (pH 7.0) and 36% (v/v) PEG 400. The crystallization of all the above proteins was carried out at 18 °C using hanging drop vapor diffusion method in 24-well plates. Crystals were harvested using nylon loops and soaked in cryoprotectant consisting of 20% glycerol and 80% mother liquor. X-ray diffraction data were collected at Shanghai Synchrotron Radiation Facility (SSRF) BL18U1 beamline. Diffraction datasets were processed using XDS (Kabsch, 2010). Data collection statistics are summarized in Table 1.

Structure determination and refinement

The structure of GAGase II was solved by selenium single-wavelength anomalous dispersion (Se-SAD) using AutoSol from Phenix suite (Adams, et al., 2011). The structure of GAGase VII was determined by molecular replacement using Phaser (McCoy, et al., 2007) with the structure of GAGase II as the search model. Initial model building was carried out using AutoBuild from Phenix suite. Then several rounds of refinement and manual building were performed alternately using Phenix.Refine and Coot (Emsley and Cowtan, 2004), respectively. The ion type was confirmed by inductively coupled plasma-mass spectrometry (ICP-MS) analysis. For each sample, 10 ml of GAGase II or GAGase VII (10 mg) was mixed with 10 ml of nitric acid in equal proportions. The mixture was then placed in a metal bath at 120 °C and heated until clarified. Subsequently, the clarified mixture was filtered through a 0.22 μ m membrane and analyzed using ICP-MS (NexION, PerkinElmer). The structures of other GAGases were predicted using RoseTTAfold (Baek, et al., 2021). Structure alignments were performed using ChimeraX with point accepted mutation (PAM)-120 matrix (Pettersen, et al., 2021). All of the structure illustrations were generated using ChimeraX (<https://www.cgl.ucsf.edu/chimerax/>) and Pymol (<https://www.pymol.org/2/>). A structure-based similarity search for GAGase II and VII were performed using the DALI server

(<http://ekhidna2.biocenter.helsinki.fi/dali/>) (Holm and Rosenstrom, 2010) . The identification of tunnels and channels in proteins were performed using the Caver Web v1.2 (<https://loschmidt.chemi.muni.cz/caverweb/>) (Stourac, et al., 2019) . The classification of identified enzymes was confirmed based on the database of carbohydrate-active enzymes (CAZy) (<http://www.cazy.org/>) (Drula, et al., 2022) and the structures of identified PLs were download from the RCSB Protein Data Bank (PDB) (<https://www.rcsb.org/>) (Berman, et al., 2000) .

Site-directed mutagenesis of GAGases

Some potentially important residues located in and around the active center were individually replaced with alanine (Ala) by using the Fast Mutagenesis Kit V2 from Vazyme Biotech Co., Ltd (Nanjing, China) and these mutants were expressed and examined for activity against various substrates (HA, CS-C, HS, and alginate) to confirm their crucial role in enzyme catalysis. The primer pairs of these mutants are shown in the *Table S2*. The expression and purification of GAGases and each variant were estimated by SDS-polyacrylamide gel electrophoresis (SDS-PAGE) followed by staining with Coomassie Brilliant Blue R-250. The results of SDS-PAGE analysis are shown in *Supplemental Figure S5*.

Molecular docking

Tertiary structure-defined GAG or alginate hexasaccharide was used to dock into the active cavity of GAGase II by using the Autodock Vina program (Eberhardt, et al., 2021; Trott and Olson, 2010) and Molecular Operating Environment. Docking by using the Autodock Vina program was performed on a search space of 15.00×18.19×19.53 Å covering the active centre (x, y, z = -2.60, -12.57, 34.13). The binding energies using non-sulfated and 4-*O*-trisulfated hexasaccharides are calculated as -5.874 and -5.163 kcal/mol, respectively.

Acknowledgements

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Wei Lin, Investigation, Validation, Visualization, Writing-Original Draft; Cao Hai-Yan, Investigation, Validation, Visualization, Writing-Original Draft; Zou Ruyi, Investigation, Validation; Du Min, Investigation, Validation; Zhang Qingdong, Investigation; Lu Danrong, Resources; Xu Xiangyu, Investigation; Xu Yingying, Investigation; Wang Wenshuang, Investigation; Chen Xiu-Lan, Investigation; Zhang Yu-Zhong, Conceptualization, Methodology, Writing-Reviewing and Editing; Li Fuchuan, Conceptualization, Methodology, Writing- Reviewing and Editing.

Data availability

All data supporting the findings of this study are available within the paper (and supporting information files). All relevant data generated during this study or analysed in this published article are available from the corresponding author on reasonable request. The atomic

coordinates and structure factors of the structures in this study have been deposited in the Protein Data Bank (PDB codes: 8KHV and 8KHW). The sequences of GAGase II (GenBank: SOD82962.1) and GAGase VII (GenBank: EDV05210.1) have already existed in NCBI database.

Additional files

Supplementary data 

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Reviewer #1 (Public review):

Summary:

This study aims to uncover molecular and structural details underlying the broad substrate specificity of glycosaminoglycan lyases belonging to a specific family (PL35). They determined the crystal structures of two such enzymes, conducted in vitro enzyme activity assays, and a thorough structure-guided mutagenesis campaign to interrogate the role of specific residues. They made progress towards achieving their aims and I appreciate the attempt of the authors to address my initial comments on the paper.

Impact on the field:

I expect this work will have limited impact on the field, although it does stand on its own as a solid piece of structure-function analysis.

Strengths:

The major strengths of the study were the combination of structure and enzyme activity assays, comprehensive structural analysis, as well as a thorough structure-guided mutagenesis campaign.

Weaknesses:

(Before revision) -the authors claim to have done a ICP-MS experiment to show Mn^{2+} binds to their enzyme, but did not present the data. The authors could have used the anomalous scattering properties of Mn^{2+} at the synchrotron to determine the presence and location of this cation (i.e. fluorescence spectra, and/or anomalous data collection at the Mn^{2+} absorption peak).

*comment after revision: I appreciate that the authors included this data now, and it looks fine.

(Before revision) -the authors have an over-reliance on molecular docking for understanding the position of substrates bound to the enzyme. The docking analysis performed was cursory at best; Autodock Vina is a fine program but more rigorous software could have been chosen, as well we molecular dynamics simulations. As well the authors do not use any substrate/product-bound structures from the broader PL enzyme family to guide the placement of the substrates in the GAGases, and interpret the molecular docking models.

*comment after revision: the authors used another docking program, which is fine, but did not do any MD analysis or comment on why not. Also maybe it is just me but I still do not see a figure explicitly showing an overlay/superposition of the docking results with crystal structures of similar enzymes with similar ligands. The authors do have a statement in this regard but I believe a figure (e.g. an additional panel on S2) would be very helpful to the reader.

(Before revision)-the conclusion that the structures of GAGase II and VII are most similar to the structures of alginate lyases (Table 2 data), and the authors' reliance on DALI, are both questioned. DALI uses a global alignment algorithm, which when used for multi-domain enzymes such as these tends to result in sub-optimal alignment of active site residues, particularly if the active site is formed between the two domains as is the case here. The authors should evaluate local alignment methods focused on optimization of the superposition of a single domain; these methods may result in a more appropriate alignment of the active site residues, and different alignment statistics. This may influence the overall

conclusion of the evolutionary history of these PL35 enzymes.

*comment after revision: I'm not sure the authors understood my suggestion as the reply reiterates the original conclusions. I suggest local structural alignment of *only* the toroid and antiparallel β -sheet domains, not global alignment of both domains, as this would improve the accuracy of the structural similarity conclusions.

(Before revision)-the data on the GAGase III residue His188 is not well interpreted; substitution of this residue clearly impacts HA and HS hydrolysis as well. The data on the impact on alginate hydrolysis is weak, which could be due to the fact that the WT enzyme has poor activity against alginate to start with.

*comment after revision: I appreciate that the authors used higher amounts of H188A variants and still do not see activity on alginate, which strengthens the conclusions regarding this substrate. However this variant also has decreased activity against HS (Figure 5C) and thus H188 appears to be important for more substrates than just alginate. The discussion section should be updated accordingly.

(Before revision)-the authors did not use the words "homology", "homologous", or "homolog" correctly (these terms mean the subjects have a known evolutionary relationship, which may or may not be known in the contexts the authors used these targets); the words "similarity" and "similar" are recommended to be used instead.

*comment after revision: I thank the authors for addressing this.

(Before revision)-the authors discuss a "shorter" cavity in GAGases, which does not make sense, and is not supported by any figure or analysis. I recommend a figure with a surface representation of the various enzymes of interest, with dimensions of the cavity labeled (as a supplemental figure). The authors also do not specifically define what subsites are in the context of this family of enzymes, nor do they specifically label or indicate the location of the subsites on the figures of the GAGase II and IV enzyme structures.

*comment after revision: I thank the authors for improving their figures and text description on this point.

<https://doi.org/10.7554/eLife.102422.2.sa2>

Reviewer #3 (Public review):

Summary:

The authors characterized previous substrate specificity of several polysaccharide lyases from family PL35 (CAzy) and discovered their unusually broad substrate specificity, being able to degrade three types of GAGs belonging to HA, CS, and HS classes.

In this study they determined the 3D structures of two lyases from this family and identified several residues essential for substrate degradation. Comparison with lyases from other PL families but having the same fold allowed them to propose an Asn, Tyr and His as essential for catalysis. One of the characterized lyases can also degrade alginate and they established a specific His residue as necessary for activity toward this substrate but not sufficient by itself. Attempts to obtain crystals with substrate or products were unsuccessful, therefore the authors resorted to modeling substrate into the determined structures. The obtained models led them to propose a catalytic mechanism, that generally reflects previously proposed mechanism for lyases with this fold.

Unfortunately, they have no definitive explanation for a broad specificity for the PL35 lyases but suggest that it is related to a shorter substrate binding cleft with a large open space on the nonreducing end of the substrate.

Strengths:

The determination of 3D structure of two PL35 lyases allows comparing them to other lyases with similar fold. The structures show a shorter substrate binding cleft that might be the reason for broader substrate specificity. Essential roles of several residues in catalysis and/or substrate binding were established by mutagenesis.

Weaknesses:

The main weakness is the lack of the structures of an enzyme-substrate/product complex. While the determined structures confirm the predicted two domain fold with a helical toroid domain and a double beta-sheet domain, the explanation for the broad specificity is lacking, except for suggestion that it has to do with a shorter substrate binding cleft. The enzymatic mechanism is hypothesized based on models rather than supported by experimentally determined structure of the complex.

<https://doi.org/10.7554/eLife.102422.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This study aims to uncover molecular and structural details underlying the broad substrate specificity of glycosaminoglycan lyases belonging to a specific family (PL35). They determined the crystal structures of two such enzymes, conducted in vitro enzyme activity assays, and a thorough structure-guided mutagenesis campaign to interrogate the role of specific residues. They made progress towards achieving their aims but I see significant holes in data that need to be determined and in the authors' analyses.

Impact on the field:

I expect this work will have a limited impact on the field, although, with additional experimental work and better analysis, this paper will be able to stand on its own as a solid piece of structure-function analysis.

Strengths:

The major strengths of the study were the combination of structure and enzyme activity assays, comprehensive structural analysis, as well as a thorough structure-guided mutagenesis campaign.

Weaknesses:

There were several weaknesses, particularly:

(1) The authors claim to have done an ICP-MS experiment to show Mn²⁺ binds to their enzyme but did not present the data. The authors could have used the anomalous scattering properties of Mn²⁺ at the synchrotron to determine the presence and location of this cation (i.e. fluorescence spectra, and/or anomalous data collection at the Mn²⁺ absorption peak).

Thank you for your kind comment and suggestion. Many studies utilized ICP-MS for the detection of metal ions within proteins (doi: 10.1016/j.jbc.2023.103047; doi: 10.1074/jbc.RA119.011790), so we utilized this method to determine the type of atoms within GAGases. In the revised manuscript, the data of ICP-MS experiment has been presented in “Supplemental Table S1”

(2) The authors have an over-reliance on molecular docking for understanding the position of substrates bound to the enzyme. The docking analysis performed was cursory at best; Autodock Vina is a fine program but more rigorous software could have been chosen, as well we molecular dynamics simulations. As well the authors do not use any substrate/product-bound structures from the broader PL enzyme family to guide the placement of the substrates in the GAGases, and interpret the molecular docking models.

Thank you for your kind comments. The interaction between the enzyme and ligand should be confirmed by resolving the structure of enzyme-ligand complex. Unfortunately, we tried to prepare the co-crystals of GAGases with various oligosaccharide substrates but ultimately failed. Thus, we tried to use docking to explain the catalytic mechanism of polysaccharide lyases using Autodock Vina although this method may be questionable. In the revised manuscript, we predicted the substrate binding site of GAGase II using Caver Web 1.2 and performed molecular docking near the substrate binding site simultaneously using Molecular Operating Environment (MOE) to verify the accuracy of the docking results (Figure 6, Supplemental Figure S4). In addition, a series of enzyme-substrate complex structures of identified PL family enzymes with structural similarities to the GAGases are showed in Supplemental Figure S2, and the positions of the catalytic cavities and the substrate binding modes are similar to those of the molecular docking results, which may also corroborate the referability of our molecular docking results in another aspect.

(3) The conclusion that the structures of GAGase II and VII are most similar to the structures of alginate lyases (Table 2 data), and the authors' reliance on DALI, are both questioned. DALI uses a global alignment algorithm, which when used for multi-domain enzymes such as these tends to result in sub-optimal alignment of active site residues, particularly if the active site is formed between the two domains as is the case here. The authors should evaluate local alignment methods focused on the optimization of the superposition of a single domain; these methods may result in a more appropriate alignment of the active site residues and different alignment statistics. This may influence the overall conclusion of the evolutionary history of these PL35 enzymes.

Thank you for your kind question. As your suggestion, multiple structural alignment assays were carried out for the $(\alpha/\alpha)_n$ toroid and the antiparallel β -sheet domain, respectively, based on the structures of GAGs/alginate lyases from PL5, PL8, PL12, PL15, PL17, PL21, PL23, PL36, PL38 and PL39 families. The results showed that the overall structure of GAGases is more similarity to that of PL15, PL17 and PL39 family alginate lyases, which have an $(\alpha/\alpha)_6$ toroid and an antiparallel β -sheet domain (Table 3). In terms of the toroid and antiparallel β -sheet domains, most of them have an $(\alpha/\alpha)_6$ toroid and an antiparallel β -sheet as shown in Table 3. We also noticed that GAGases possess such a $(\alpha/\alpha)_6$ toroid structure rather than a $(\alpha/\alpha)_7$ toroid structure, and revised the relevant statement in the manuscript.

(4) The data on the GAGase III residue His188 is not well interpreted; substitution of this residue clearly impacts HA and HS hydrolysis as well. The data on the impact on alginate hydrolysis is weak, which could be due to the fact that the WT enzyme has poor activity against alginate to start with.

Thank you very much for your helpful comments and questions. To verify your suggestion that the weak impact of alginate hydrolysis could be due to poor activity of wild type GAGase III, we degraded alginate using different enzyme concentrations (3 to 30 μ g) and analyzed the degradation products. The results showed that the alginate-degrading activity of GAGase III-H188A and GAGase III-H188N was abolished, even at a quite high ratio of the mutated enzyme to substrate such as 30 μ g enzyme to 30 μ g substrate (Supplemental Figure S3A), while their GAG-degrading activity was only partially affected, indicating that this residue plays a more important role for the digestion of alginate than other substrates. Unfortunately, we were unable to confer the ability to GAGase III through the mutation of N191H in GAGase II. Therefore, we suggest that His¹⁸⁸ play a key role in the specificity of alginate degradation by GAGase III, but that other determinants also contribute to this process. We will try more methods to obtain the structure of enzyme-substrate co-crystals and explain its substrate-selective mechanism in future studies.

(5) The authors did not use the words "homology", "homologous", or "homolog" correctly (these terms mean the subjects have a known evolutionary relationship, which may or may not be known in the contexts the authors used these targets); the words "similarity" and "similar" are recommended to be used instead.

Thank you for your helpful suggestions. We have revised the relevant part of the description in the manuscript.

(6) The authors discuss a "shorter" cavity in GAGases, which does not make sense and is not supported by any figure or analysis. I recommend a figure with a surface representation of the various enzymes of interest, with dimensions of the cavity labeled (as a supplemental figure). The authors also do not specifically define what subsites are in the context of this family of enzymes, nor do they specifically label or indicate the location of the subsites on the figures of the GAGase II and IV enzyme structures.

Thank you for your helpful suggestions. Figures (Supplemental Figure S2) with surface representations of the GAGase II and some structurally similar GAGs/alginate lyases with the dimensions of the cavity labeled, were added to the supplementary data as you suggested. Considering the correlation between enzyme specificity and substrate binding sites, we speculated that a shorter substrate binding cavity might allow the enzyme to accommodate a wider variety of substrates, resulting in a smaller restriction of the catalytic cavity to substrate binding, although this speculation needs to be verified by the resolution of the crystal structure of the enzyme-substrate complexes.

Reviewer #2 (Public review):

Summary:

Wei et al. present the X-ray crystallographic structures of two PL35 family glycosaminoglycan (GAG) lyases that display a broad substrate specificity. The structural data show that there is a high degree of structural homology between these enzymes and GAGases that have previously been structurally characterized. Central to this are the N-terminal (α/α)₇ toroid domain and the C-terminal two-layered β -sheet domain. Structural alignment of these novel PL35 lyases with previously deposited structures shows a highly conserved triplet of residues at the heart of the active sites. Docking studies identified potentially important residues for substrate binding and turnover, and subsequent site-directed mutagenesis paired with enzymatic assays confirmed the importance of many of these residues. A third PL35 GAGase that is able to turn over alginate was not crystallized, but a predicted model showed a conserved active site Asn was mutated to a His, which could potentially explain its ability to act on alginate.

Mutation of the His into either Ala or Asn abrogated its activity on alginate, providing supporting evidence for the importance of the His. Finally, a catalytic mechanism is proposed for the activity of the PL35 lyases. Overall, the authors used an appropriate set of methods to investigate their claims, and the data largely support their conclusions. These results will likely provide a platform for further studies into the broad substrate specificity of PL35 lyases, as well as for studies into the evolutionary origins of these unique enzymes

Strengths:

The crystallographic data are of very high quality, and the use of modern structural prediction tools to allow for comparison of GAGase III to GAGase II/GAGase VII was nice to see. The authors were comprehensive in their comparison of the PL35 lyases to those in other families. The use of molecular docking to identify key residues and the use of site-directed mutagenesis to investigate substrate specificity was good, especially going the extra distance to mutate the conserved Asn to His in GAGase II and GAGase VII.

Weaknesses:

The structural models simply are not complete. A cursory look at the electron density and the models show that there are many positive density peaks that have not had anything modelled into them. The electron density also does not support the placement of a Mn²⁺ in the model. The authors indicate that ICP-MS was done to identify the metal, but no ICP-MS data is presented in the main text or supplementary. I believe the authors put too much emphasis on the possibility of GAGase III representing an evolutionary intermediate between GAG lyases and alginate lyases based on a single Asn to His mutation in the active site, and I don't believe that enough time was spent discussing how this "more open and shorter" catalytic cavity would necessarily mean that the enzyme could accommodate a broader set of substrates. Finally, the proposed mechanism does not bring the enzyme back to its starting state.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Minor points:

(1) The number of significant digits used in Table 1 and Figure 3 legend are not justified. The authors should use a maximum of 2 significant digits.

Thank you for your kind suggestion. We have verified the relevant data and retained two significant digits.

(2) The authors should use the words "mutant" or "mutation" only when discussing DNA, but when discussing protein, the words "variant" and "substitution" should be used instead as these are more appropriate.

Thank you for your helpful suggestions. We have revised the relevant description in the manuscript as you suggested.

(3) Lines 102-110 are a long, run-on sentence that should be split into shorter sentences. Similarly, lines 367-378 should be split into shorter sentences.

Thank you for your suggestions. In the revised manuscript, the long sentences in lines 102-110 and 367-378 have been rewritten into shorter ones.

(4) Lines 174-175: *His, Tyr, Glu, and Trp are not positively charged residues and this wording should be changed.*

Thank you for your suggestions. We have revised the relevant description in the manuscript as you suggested.

(5) Lines 423-426 *require a reference.*

Thank you for your suggestion. We have provided the reference at the right position and revised the relevant description in the manuscript as you suggested.

(6) *Grammar/language:*

-line 90 - change "should emerge" to "likely emerged"

-line 145 - delete "Finally"

-line 264 - delete "their"

-line 265 - delete "active sites"

-line 265-266 - change to "To confirm this hypothesis, site-directed mutagenesis followed by enzyme activity assay was performed"

-line 311 - change "residue in the catalytic cavity of GAGase III, which.." to "residue in its catalytic cavity, which..."

-line 318 - change "affect" to "affected"

-line 323 - change to "degrading activity of GAGase II remains to be determined outside of the His188 residue"

-line 345 - delete "assays"

-line 359 - change to "evidence"

-line 397 - change "folds" to "3D fold"

-line 420 - change to "share similar catalytic sites"

-lines 411, 433 - change "conversed" to "conserved"

-line 441 - change to "Mutational analysis showed that the His188.."

-line 450 - delete "which"

Thank you for your suggestions. Grammatical errors in the revised manuscript have been corrected in the revised manuscript.

Reviewer #2 (Recommendations for the authors):

Major Concerns

The electron density in your model clearly does not support the placement of a Mn ion. In the GAGase II structure, the placement of the Mn and the placement of waters around it still results in two density peaks of > 12 rmsd. The manuscript suggests that ICP-MS was done but the results of this are not shown anywhere. Please include your ICP-MS data. I see the structures have already been deposited, and if they have been deposited

unchanged, please see if you can modify them to actually finish building the models. I don't find your data in Figure 2B particularly convincing that Mn is necessarily important for activity.

Thank you for your kind comments. As we known, ICP-MS is a common method used for the detection of metal ions within proteins (doi: 10.1016/j.jbc.2023.103047; doi: 10.1074/jbc.RA119.011790), and thus we utilized it to determine the type of atoms within GAGases in this study. In the revised manuscript, the data of ICP-MS experiment has been presented in “Supplemental Table S1”, and the data clearly showed that the content of Mn^{2+} rather than others in test sample is much higher than that in the negative control, suggesting the involvement of Mn^{2+} in the protein. We agree that the addition of Mn^{2+} does not show very strong promotion to the activity of GAGase II just like other tested metal ions, but the addition of EDTA significantly inhibited the enzyme activity (Figure 2), indicating that metal ion such as Mn^{2+} is necessary for the function of GAGases. Regarding the role of metal ion, whether it participates in the catalytic reaction or only stabilize the structure of enzyme remains to be further explored in our further study.

Minor Concerns

(1) Please include CC1/2 in your Table 1.

Thank you for your kind suggestions. CC1/2 parameters have been added in the revised manuscript (Table 1).

(2) If possible please include SDS-PAGE gel images of your purified proteins. Particularly for the point mutations. Ideally, you would have done SEC on your mutants to show that the reduction in activity is not due to aggregation/misfolding, but at the very least I would to see that you have similar levels of purity.

Thank you for your kind suggestions. As your suggestion, we have added SDS-PAGE gel images of purified GAGase II, GAGase III, GAGase VII, and their mutant enzymes to the supplementary data. As shown in Figure S5, site-directed mutagenesis did not affect the soluble expression levels of GAGase II, GAGase III or GAGase VII, indicating that the reduction in activity is not due to aggregation or misfolding. Due to the large number of variants, we used crude enzyme for the activity assay of substrate binding sites, while for some catalytic key residues, we purified the corresponding mutant enzymes and then verified their activities by HPLC.

(3) When referring to your structural predictions, it is not appropriate to say that you used Robetta. Your reference is correct though - you should say that the structures were predicted using RoseTTAfold.

Thank you for your helpful suggestions. We have revised the relevant description in the manuscript.

(4) If possible expand on how the shorter/more open active site cavity would result in broader substrate specificity.

Thank you for your kind comment. In the revised manuscript, figures (Supplemental Figure S2) with surface representations of the GAGase II and some representatively structurally similar GAGs/alginate lyases, with the dimensions of the cavity labeled, were added to the supplementary data. Considering the correlation between enzyme specificity and substrate binding sites, we speculated that a shorter substrate binding cavity might allow the enzyme to accommodate a wider variety of substrates, resulting in a smaller restriction of the

catalytic cavity to substrate binding. However, unfortunately, we did not succeed in obtaining co-crystals of GAGases with any of the substrates. We will try to explain the mechanism of substrate selectivity in future studies by culturing and resolving crystals of its enzyme substrate complex or otherwise.

(5) *I would put less emphasis on His188 in GAGase III being a strong indicator that this protein represents an evolutionary intermediate between alginate lyases and GAGases.*

Thank you for your comment. The His¹⁸⁸ residue, which is unique compared to other GAGases, is essential for the alginate-degrading activity of GAGase III. Regarding why GAGases are thought to represent a possible evolutionary intermediate between alginate lyases and GAG lyases, phylogenetic analysis demonstrated that GAGases show considerable homology with some identified GAG lyases and alginate lyases (DOI: 10.1016/j.jbc.2024.107466). The similarity in primary structure between some GAG lyases, alginate lyases, and GAGases suggests structural similarities, which are further supported by this study. As structure determines function, structural similarity is often used as a key criterion when studying the evolution of proteins, the GAGase III, which shows significant GAGs and alginate-degrading activity, support for this speculation. Of course, in this study, our analysis of the evolutionary relationship between GAGases and identified GAG lyases and alginate lyases, based on structural comparison, is an attempt using existing methods. The conclusions we have drawn remain a hypothesis that still requires further evidence to support and validate.

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